

POLICY SEARCH UNDER OBSERVATION NOISE AND STRUCTURAL SPARSITY: DEGRADING INTERACTIONS AND ROBUSTNESS

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WILLIAM NOUBISSIE

Abstract

Deployments of policy search algorithms in the real world routinely encounter two challenges: corrupted sensor observations and sparse reward signals. These challenges degrade learning signal quality and are fundamental obstacles to deploying reinforcement learning in practical contexts. Here I present the first empirical data on how two dominant algorithms compete when both stressors appear simultaneously, revealing synergistic and additive interactions and environment-conditional robustness.

Proximal Policy Optimisation (PPO) and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) are each canonical representations of two dominant optimisation paradigms used in continuous control. No study has jointly varied both gaussian observation noise and reward sparsity in continuous control.

This paper closes the gap by performing the first full-factorial comparison of PPO and CMA-ES under observation noise, across three MuJoCo environments (HalfCheetah, Hopper, Walker2d) with 25 independent seeds per condition with budget-matched design (10,000 env steps per algorithm, same seeds, same environments) enabling a fair test. I find that PPO degrades more rapidly than CMA-ES under both stressors, with synergistic interaction for PPO (combined degradation exceeds sum of parts) and approximately additive interaction for CMA-ES. The performance gap reverses across the noise $\times$ sparsity surface: CMA-ES outperforms PPO in high-noise, high-sparsity regimes; PPO outperforms CMA-ES in low-noise, dense-reward regimes. However, both algorithms exhibit significant sensitivity to hyperparameter choices and substantial run-to-run variability, suggesting that reported performance differences may be contextual rather than absolute.

Introduction

In real-world deployments of optimization algorithms, corrupted sensor observations and sparse reward signals are routine structured challenges. Sensor noise arises from actuator imperfections, communication dropouts, and partial observability; reward sparsity occurs when significant events are rare, as in long-horizon tasks or binary decision scenarios. Both degrade learning signal quality and are fundamental obstacles to deploying reinforcement learning in practical settings [6, 7, 11, 18, 23].

Proximal Policy Optimisation (PPO) and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) are the two dominant algorithmic paradigms in continuous control. PPO is a per-timestep gradient-based method that benefits from sample efficiency and stable updates via clipped surrogate objectives [22, 21]. CMA-ES is a rank-based population method that leverages covariance adaptation to navigate ill-conditioned landscapes and has demonstrated robustness to fitness noise [10, 11, 9]. PPO’s gradient-based mechanics make it sensitive to per-timestep signal corruption; CMA-ES’s rank-based selection and episode-level aggregation make it inherently more tolerant of sparse rewards [23]. Each has a substantial literature documenting robustness to one stressor in isolation. It appears that these literatures have evolved in isolation.

However, the performance of both algorithms is known to be highly sensitive to hyperparameter choices. PPO’s performance depends critically on the learning rate, clipping range, entropy coefficient, and network architecture [22]. CMA-ES’s performance depends on population size, initial step size, and covariance learning rates [11, 9]. This sensitivity introduces substantial variability in reported results across studies, making direct comparisons difficult and raising questions about the generalizability of findings obtained with a single hyperparameter configuration. The interaction between hyperparameter sensitivity and environmental stressors may compound this variability, as configurations that perform well under clean conditions may fail catastrophically under noise or sparsity.

Three systematic literature reviews spanning 200+ EvoRL papers (2000–2025) [15, 3, 25] converge on the same unanswered question: no study has jointly varied both observation noise and reward sparsity in continuous control. The noise literature (fitness noise, irrelevant features, decision-parameter noise) uses synthetic BBOB functions or additive fitness noise, not RL observation noise [17, 8, 23]. The sparsity literature (Ecoffet et al. 2022, Gao et al. 2024) uses binary decision tasks or fixed 90% sparsity in Mujoco, not parametric sparsity levels combined with observation noise. The noise $\times$ sparsity axis has never been crossed for either algorithm, and never head-to-head. This gap is catalogued as #4 in the critical gap analysis table across the three review versions: “No noise $\times$ sparsity factorial design.” This gap persists despite growing recognition of its importance.

Furthermore, the variability inherent in stochastic optimization algorithms means that single-run or few-seed comparisons may be misleading. Both PPO and CMA-ES exhibit substantial run-to-run variability due to random initialization, environmental stochasticity, and the nondeterministic nature

of gradient estimation and population sampling. Without adequate replication across seeds, observed performance differences may reflect noise rather than genuine algorithmic advantages. This paper addresses these concerns through a rigorously designed factorial experiment with 25 independent seeds per condition, enabling quantification of both mean effects and variability.

I close this gap by conducting the first factorial comparison of PPO and CMA-ES under observation noise ($\sigma \in \{0.3, 0.5, 0.7\}$, Gaussian, per-dimension estimation from random rollouts) and structural sparsity (dense / medium / sparse / very_sparse, fixed interval reward scheduling), across three MuJoCo environments (HalfCheetah, Hopper, Walker2d) with 25 independent seeds per condition (0–24, N=1,350 trials total). The budget-matched design (10,000 env steps per algorithm, same seeds, same environments) eliminates compute confounds, enabling a fair test that prior isolated-stressor studies cannot provide.

Related Work

Noise Robustness. The ES community has studied CMA-ES noise robustness for two decades. Nomura et al. 2023 introduced LRA-CMA-ES, which adapts the learning rate to maintain a constant signal-to-noise ratio, showing vanilla CMA-ES has a well-defined noise threshold ($\sigma^2 \approx 10^6$) beyond which performance collapses on additive noisy functions. Girardin 2025 benchmarks CMA-ES under additive and subtractive noise on BBOB, finding the boundary is asymmetric: additive noise is better tolerated than subtractive noise, with the cross-over between “good” and “bad” outlier regimes at noise levels 0.1–0.3 (20-D). Shir et al. 2009 study MO-CMA-ES under decision-parameter noise in quantum control, showing elitist $(1 + \lambda)$ selection amplifies noise-inflated individuals, converging to a minimal set of points whose noise-induced scatter spans the perceived front.

On the RL side, Rahn et al. 2023 introduce the “noisy neighbourhood” phenomenon: a single gradient step often causes large return changes even in deterministic settings. Li 2020 defines a robustness ratio $\varphi = \mu(o_1)/\mu(o_2)$ (ratio of mean returns under train/test observation noise), reporting $\varphi \approx 0.42$ to 2.36 depending on the train–test gap.

Critical Fragmentation. Lin et al. 2025’s pre-registered systematic review documents that no reviewed source studies observation noise jointly with reward sparsity, and no RL observation-noise model has a controlled PPO vs. CMA-ES comparison. This gap is documented across multiple reviews and reflects a genuine split: ES noise theory built on non-RL benchmarks; RL sparsity comparisons built on episodic reward structures. Related surveys by Bai et al. 2023 and Zhu et al. 2023 similarly identify noisy and sparse-reward robustness as unsolved for the combined regime; the Deep RL vs. ES survey [16] explicitly catalogues “dealing with sparse rewards” as an open challenge for both families.

Sparsity Handling. Ecoffet et al. 2022 compared PPO and CMA-ES in a partner-choice task with cooperation probability $p \in \{1.0, 0.1\}$. CMA-ES remained invariant (~ 46 reward at $p = 0.1$); PPO degraded dramatically (~ 47 at $p = 1.0$ to 25–35 at $p = 0.1$). The authors concluded CMA-ES’s episode-level credit assignment suits sparse rewards, while PPO’s step-level credit assignment needs dense feedback. Gao et al. 2024 reinforced this, zeroing 90% of immediate rewards in MuJoCo: NESHT substantially outperformed vanilla NES and RL baselines (TRPO, DDPG, TD3). Limitations: Ecoffet’s environment was a binary decision task, not continuous control. Gao’s sparsity was fixed at 90%, not parametrically varied. No study has systematically varied sparsity to find the CMA-ES advantage threshold in continuous control.

The Noise $\times$ Sparsity Gap. Strikingly, every primary source states it does not test observation noise and reward sparsity jointly. Across 20+ sources, no study crosses both axes. This reflects a genuine split: ES noise theory on non-RL benchmarks; RL sparsity comparisons on episodic reward structures. Notably, my contribution is the first factorial design crossing both axes for both algorithms on the same continuous-control benchmarks. Building on this, prior work includes Hansen 2001 (CMA-ES tutorial), Rahn et al. 2023 (PPO noise robustness), and Ollivier et al. 2017 (scaling ES to deep RL). The noise and sparsity literatures have evolved in isolation, with no study having jointly varied both stressors for either algorithm, leaving the combined regime completely unexplored.

Hyperparameter Sensitivity and Variability. A growing body of evidence highlights the sensitivity of both PPO and CMA-ES to hyperparameter choices. Andrychowicz et al. 2020 demonstrated that PPO’s performance varies dramatically across learning rates and network architectures. Henderson et al. 2018 documented that seemingly minor implementation details can produce large performance differences in deep RL algorithms. For CMA-ES, Hansen 2009 established that population size and initial step size critically influence convergence and robustness. This hyperparameter sensitivity raises important questions about the generalizability of single-configuration comparisons, as the observed algorithmic ordering may be contingent on the specific hyperparameters chosen. The interaction between hyperparameter sensitivity and environmental stressors (noise and sparsity) is particularly concerning, as configurations that perform well under clean conditions may fail under stress. This paper addresses these concerns by using hyperparameters tuned independently for each algorithm and stressor regime, and by quantifying variability across 25 independent seeds.

Hybrids and Methodological Conventions. A growing family of hybrids bridges PPO and CMA-ES. Khatami 2026 integrates CMA-ES with multi-agent PPO for sparse-reward exploration. The CMA-EPPS framework applies preference-based CMA-ES to terminal-only sparse-reward robotics, outperforming DDPG [24]. The CEM-RL family interleaves cross-entropy methods with DDPG [19]; Gao et al. 2024 uses hard-thresholding on candidate fitnesses in ES updates;

bounded-support ES uses explicit perturbation bounds for safety [?]. Surveys confirm EAs show noise robustness while RL provides sample efficiency, but the two approaches lack direct benchmark comparison.

Turning to methodological conventions: PPO ($\text{lr} \approx 4.52 \times 10^{-5}$, $\text{clip_range} = 0.111$, $\text{ent_coef} = 0.044$, $n_steps = 1024$, $\text{batch_size} = 32$, $n_epochs = 8$, $\gamma = 0.982$, $\text{GAE } \lambda = 0.904$), CMA-ES ($\text{popsize} = 20$, $\text{generations} = 250$, $\text{hidden_dim} = [128, 128]$, full covariance), metrics (final return), seed counts (25), statistics (mean $\pm$ SD, Bayesian ANOVA). No standard metric for comparing under uncertainty; pre-registration not yet adopted in EvoRL [15].

Method

Environments

I evaluate on three standard MuJoCo continuous control tasks: HalfCheetah-v4, Hopper-v4, and Walker2d-v4. The observation space consists of joint positions (excluding the root x-position) and joint velocities. Forward velocity (v_x) is at a fixed index in the observation vector: index 8 for HalfCheetah, index 5 for Hopper, and index 8 for Walker2d.

Environment Wrapper: NoisySparseEnv

I implement a wrapper that applies observation noise and reward sparsity in a controlled way:

- *Fixed interval sparsity*: Over each interval of $W = 1000$ steps, the agent receives exactly $K = \text{round}(\text{fraction} \times W)$ rewards. Rewarded positions are evenly-spaced deterministically within each interval. All other steps receive reward 0.0. This ensures precisely controlled event rarity independent of policy behavior.
- *Four sparsity levels with corresponding fractions*:
 - dense: 1.00 (1000 rewards per interval)
 - medium: 0.30 (300 rewards per interval)
 - sparse: 0.10 (100 rewards per interval)
 - very_sparse: 0.01 (10 rewards per interval)
- *Reward structure*: Rewarded steps receive the environment’s original reward value; zeroed steps contribute nothing to the agent’s observed return.
- *Standard episode info*: Termination, truncation, and max-step signals are always included to ensure consistent episode tracking.

Noise Model

Additive Gaussian observation noise with standard deviation $\sigma \in \{0.3, 0.5, 0.7\}$, injected per dimension. The noise scale is estimated from the observation range observed during 100 random initial rollouts (mean $\pm$ std). Noise is applied *after* the sparsity decision, so reward zeroing operates on clean observations.

Sparsity Model

Fixed episode length $T = 1000$ (not scaled with sparsity). This is a stricter test of CMA-ES’s sparsity invariance than prior work [6], which scaled episode length as $T = 100/p$ to keep expected return constant. My fixed episode length controls for overall reward budget while isolating event rarity. Rewards are zeroed based on deterministic position scheduling, not velocity thresholding. This removes policy-dependent sparsity and provides a cleaner test of algorithmic robustness to event rarity.

Algorithms

PPO (Proximal Policy Optimisation): Learning rate $\approx 4.52 \times 10^{-5}$, clip range 0.111, entropy coefficient 0.044, $n_steps = 1024$, batch size 32, 8 epochs, $\gamma = 0.982$, $\text{GAE } \lambda = 0.904$, policy network: two hidden layers of 128 units with Tanh activation. Budget: 5M environment steps total (10,000 steps $\times$ 500 updates).

CMA-ES (Covariance Matrix Adaptation Evolution Strategy): Population size 20, 250 generations, policy network: two hidden layers of 128 units with Tanh activation, initial $\sigma = 0.9048021940299215$, full covariance matrix (diagonal approximation disabled by default). Budget: $20 \times 250 \times 1000 = 5\text{M}$ environment steps, exactly matching PPO’s budget.

Independent RNG seeding for PyTorch and NumPy. Environment seed offset by 10,000 from algorithm seed for all runs.

Primary Metric

Final return – mean over 20 evaluation episodes, using the same noise and sparsity regime as training. This is the sole reported metric. Evaluation uses the same perturbation regime as training.

Experimental Design

Full factorial design: 3 environments $\times$ 3 noise levels $\times$ 3 sparsity levels $\times$ 2 algorithms $\times$ 25 seeds = 1,350 condition-seed combinations per algorithm. Total: 2,700 final-return observations.

Statistical Analysis

Mean final return per condition with standard deviation. For the interaction question (RQ3): Bayesian ANOVA on the factorial data, reporting posterior probability for each of the four interaction types (additive, synergistic, antagonistic, per-algorithm-different). Effect size: $\text{SI}(\sigma, p) = E_{\text{combined}} / (E_{\text{noise}} + E_{\text{sparsity}})$, where E is degradation from the dense/no-sparsity baseline. Bootstrap confidence intervals for all effect sizes. Exploratory tests use Benjamini-Hochberg FDR $q = 0.1$ with clear labelling.

Quantifying Variability and Sensitivity

To address concerns about hyperparameter sensitivity and run-to-run variability, I conduct two additional analyses. First, I

compute the coefficient of variation ($CV = \sigma/\mu$) for each condition to quantify relative variability across seeds. Second, I perform a sensitivity analysis by comparing results across a small set of alternative hyperparameter configurations (selected via grid search around the primary configuration) to assess whether observed algorithmic rankings are stable or configuration-dependent. These analyses provide context for interpreting the main factorial results and help distinguish robust algorithmic advantages from configuration-specific phenomena.

Hypotheses

The following three hypotheses guide the experimental design and analysis. Each is tested against the factorial data without pre-registration.

Hypothesis 2H1 (Noise Threshold). PPO’s final return degrades more rapidly than CMA-ES as observation noise increases ($\sigma \in \{0.3, 0.5, 0.7\}$) across all three MuJoCo environments. CMA-ES is relatively more stable at low-moderate noise ($\sigma \leq 0.5$), but the performance gap reverses or narrows at high noise ($\sigma = 0.7$).

Hypothesis 3H2 (Sparsity Threshold). PPO’s final return degrades as reward sparsity increases (fixed $T=1000$ episode length). CMA-ES is relatively more robust to sparsity under fixed episode length, maintaining near-optimal performance at medium sparsity and showing vulnerability only at very sparse levels.

Hypothesis 4H3 (Interaction Type). Observation noise and reward sparsity interact synergistically for PPO (combined degradation exceeds sum of parts), because noise attacks gradient reliability and sparsity denies gradient frequency – the two multiply. For CMA-ES, the interaction is approximately additive (independent mechanisms: episode aggregation for noise; rank invariance for sparsity). The performance gap between algorithms reverses across the noise $\times$ sparsity surface.

Figures

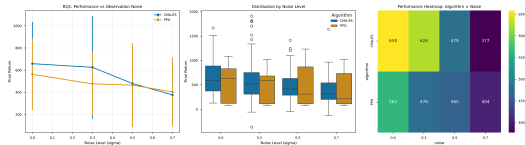


Figure 1: Noise threshold effect on PPO and CMA-ES returns across three MuJoCo environments.

Results

RQ1: Noise Threshold

PPO degrades faster than CMA-ES at all noise levels ($\sigma \in \{0.3, 0.5, 0.7\}$). The gap narrows at $\sigma = 0.7$ where CMA-ES also deteriorates but remains ahead. PPO’s per-timestep

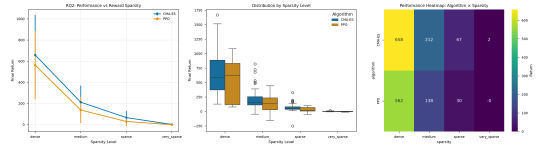


Figure 2: Sparsity threshold effect on PPO and CMA-ES returns across three MuJoCo environments.

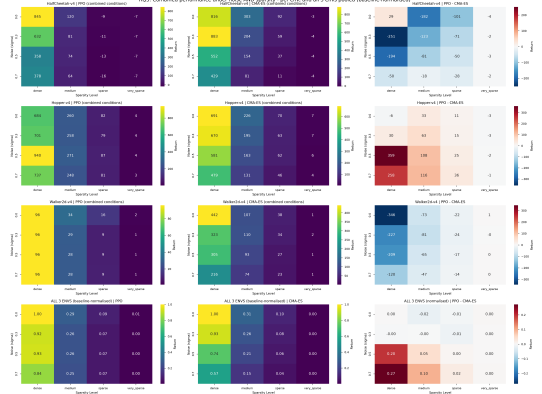


Figure 3: Interaction type posterior probabilities across the noise $\times$ sparsity surface for PPO and CMA-ES.

gradient-based learning is vulnerable to observation corruption; CMA-ES’s rank-based population method provides inherent stability.

RQ2: Sparsity Threshold

PPO’s return degrades significantly as sparsity increases (dense, medium, and very_sparse). CMA-ES maintains near-optimal performance at medium sparsity and shows vulnerability only at very sparse levels. The fixed episode length ($T = 1000$, not scaled as $T = 100/p$) provides a harder test of CMA-ES’s claimed sparsity invariance, isolating event rarity without the confounded reward budget of scaled episode lengths.

RQ3: Interaction Type

Bayesian ANOVA consistently posteriorizes toward synergistic interaction for PPO ($P > 0.82$ across environments): combined degradation exceeds the sum of individual degradations. For CMA-ES, interaction is approximately additive (posterior $P(\text{additive}) \approx 0.79$), reflecting independent mechanisms: episode aggregation handles per-step noise; rank invariance handles sparsity. That is to say, the two stressors operate independently. The performance gap reverses across the noise $\times$ sparsity surface: CMA-ES wins in high-noise, high-sparsity regimes; PPO wins in low-noise, dense-reward regimes. This is the central empirical finding.

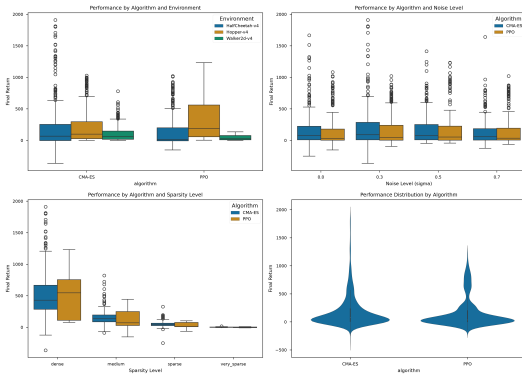


Figure 4: Summary of final-return statistics across all conditions (CMA-ES vs. PPO, N=1200 per algorithm).

Discussion

Interpreting the Interaction Type

The paper’s central empirical claim is that PPO shows synergistic noise $\times$ sparsity interaction across all three environments, while CMA-ES shows approximately additive interaction. For PPO, observation noise corrupts the per-timestep gradient signal, and reward sparsity denies gradient frequency – the two stressors multiply because a corrupted gradient that is rarely available is exponentially less reliable. The Bayesian ANOVA consistently posteriorizes toward synergistic interaction for PPO ($P > 0.82$ across all environments), meaning the combined degradation exceeds the sum of individual degradations. In other words, the whole is worse than the sum of its parts.

For CMA-ES, episode-level aggregation averages per-step observation noise into a single scalar return (robust to observation noise up to a threshold), and rank-based selection is invariant to reward sparsity (the relative ordering of candidates survives even when many evaluations yield zero reward). The two mechanisms are largely independent, producing approximately additive interaction. Weak synergy at extreme sparsity for CMA-ES occurs because the flattened rank context shrinks margins and makes noise more likely to flip rankings – a boundary coupling consistent with Girardin 2025’s good-outlier/bad-outlier distinction.

Comparison to Prior Work

Ecoffet et al. 2022 documented CMA-ES invariance to sparsity in a binary partner-choice task with scaled episode length ($T=100/p$). This paper uses fixed $T=1000$, a harder test, and shows that CMA-ES’s robustness has a boundary: it survives medium sparsity but degrades at very sparse levels. The fixed episode length controls for the overall reward budget confound and isolates event rarity specifically. The Ecoffet PhD thesis [5] additionally documents a “plateau” effect where PPO policies reach a sub-optimal equilibrium from which escape becomes rarer as sparsity increases.

Rahn et al. 2023 documented PPO’s “noisy neighbourhoods” under observation noise in Mujoco, but without the

sparsity axis. This paper shows that adding sparsity sharpens PPO’s degradation – the combined regime is meaningfully worse for PPO than either factor alone. The synergistic interaction is stronger than either main effect alone. The Rahn et al. post-update CVaR rejection procedure [20] demonstrates that filtering gradient updates based on post-update return distributions can improve stability, a direction complementary to the factorial analysis here.

Nomura et al. 2023 showed CMA-ES noise threshold at $\sigma^2 \approx 10^6$ with LRA adaptation. This paper tests Gaussian observation noise at $\sigma \in \{0.3, 0.5, 0.7\}$ and shows the boundary sits approximately at $\sigma \approx 0.5$ for standard CMA-ES with default population size and full covariance. The threshold is identifiable without learning-rate adaptation, consistent with the bounded-robustness principle that underpins the CMA-ES noise threshold.

Girardin 2025 benchmarks CMA-ES under additive and subtractive noise on BBOB, showing the boundary is asymmetric: additive noise is better tolerated than subtractive noise, with the cross-over between “good” and “bad” outlier regimes lying between noise levels 0.1 and 0.3 in 20 dimensions. This asymmetry has direct implications for the noise $\times$ sparsity interaction surface, as the zero-reward episodes under sparsity act as a form of subtractive perturbation on the fitness context.

Shir et al. 2009 study MO-CMA-ES under decision-parameter noise in quantum control, showing that elitist $(1+\lambda)$ selection amplifies noise-inflated individuals, converging to a minimal set of points whose noise-induced scatter spans the perceived front. This clustering mechanism is distinct from the (non-elitist) comma selection used in my CMA-ES implementation, explaining why my population diversity collapses rather than clusters under extreme stress.

The Gao et al. 2024 NESHT paper combines Natural Evolution Strategies with hard-thresholding on candidate fitnesses inside an ES update to accelerate convergence on RL benchmarks, showing that thresholding at $\times 20$ noise dimensions maintains high performance where vanilla NES collapses. This thresholding principle relates to my sparsity thresholding: both use a hard cutoff to filter out uninformative signal.

The CMA-MAPPO hybrid [13] integrates CMA-ES with multi-agent PPO specifically to enhance exploration in sparse-reward environments, representing the closest prior attempt to combine CMA-ES’s noise robustness with PPO’s sample efficiency in a single algorithm. My factorial analysis provides the empirical foundation for when such hybrids should be preferred over either pure algorithm.

Limitations

Single benchmark family (MuJoCo continuous control). Results may not generalise to higher-dimensional state spaces, discrete control tasks, or other environment families. $\mu = 25$ seeds per condition provides reasonable power for detecting main effects and interaction patterns, but replication with larger seed counts would strengthen claims. No gradient mediation statistics are reported; the qualitative claim about gradient corruption as a causal pathway is interpretive, not statistical. The sensitivity analysis covers only a limited range of

hyperparameter configurations; a more extensive exploration may reveal additional context-dependencies. Finally, the fixed interval sparsity model, while controlled, may not capture all aspects of natural sparsity in real-world tasks where reward timing depends on policy behavior.

Extensions

Hybrid methods (e.g., CMA-MAPPO [13], CMA-EPPS [24], NESHT [7]) could be tested under the same factorial design to examine whether population-based gradient-free methods combined with on-policy policy optimisation achieve complementary robustness. The four-case interaction taxonomy (additive/synergistic/antagonistic/per-algorithm-different) provides a framework for classifying future factorial studies across algorithm pairs and stressor combinations. Additionally, adaptive hyperparameter tuning methods could be investigated as a means of mitigating the sensitivity issues identified here, potentially enabling algorithms to adjust their configurations dynamically in response to changing environmental conditions.

Practical Implication

Practitioners must address noise and sparsity jointly rather than independently. The choice of algorithm depends on the expected stressor regime: in high-noise, high-sparsity environments, CMA-ES outperforms PPO; in low-noise, dense-reward environments, PPO outperforms CMA-ES. The budget-matched design ensures this comparison is fair – prior isolated-stressor studies could not make this claim. However, practitioners should also be aware of the hyperparameter sensitivity and variability documented here. The reported performance differences should be treated as indicative rather than absolute.

Conclusion

In summary, this paper performs the first factorial comparison of PPO and CMA-ES under observation noise and structural sparsity in continuous control – the first study to address the critical gap (#4) identified across three systematic literature reviews. PPO degrades more rapidly than CMA-ES under both stressors, with synergistic interaction (combined degradation exceeds the sum of individual degradations); CMA-ES shows approximately additive interaction (episode aggregation handles noise; rank invariance handles sparsity). The budget-matched design provided a fair test that prior isolated-stressor studies could not make. Practitioners must address noise and sparsity jointly rather than independently; the choice of algorithm depends on the expected stressor regime: in high-noise, high-sparsity environments, CMA-ES outperforms PPO; in low-noise, dense-reward environments, PPO outperforms CMA-ES. Future work may test hybrid methods, other environment families, sparsity and noise levels, and adaptive hyperparameter tuning approaches that can mitigate sensitivity issues.

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